0104. Maximum Depth of Binary Tree

y | Category: Trees | LeetCode: https://leetcode.com/problems/maximum-depth-c

Problem

- Return the number of nodes along the longest root-to-leaf path.

Intuition

- Depth(node) = 1 + max(depth(left), depth(right)), with depth(null) = 0.

Brute Force

- Enumerate all root-to-leaf paths and take the longest - same O(n), more bookkeeping.

Optimal Approach

- Recursive DFS as above, or
- BFS level by level and count levels.

Worked Example

```
[3,9,20,null,null,15,7].

depth(9)=1  depth(15)=1  depth(7)=1
depth(20)=1+max(1,1)=2
depth(3)=1+max(1,2)=3
```

Java Solution

```java
public class MaximumDepthOfBinaryTree {
    public int maxDepth(TreeNode root) {
        if (root == null) return 0;
        return 1 + Math.max(maxDepth(root.left), maxDepth(root.right));
    }
}
```

Complexity

- Time: O(n). Space: O(h).

Pitfalls

- Depth counts nodes, not edges, here.